

Predicting Weeks on Billboard for Rolling Stone's Top 500 Album

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Summary

Here, we attempt to build a regression model using the k-nearest neighbors (kNN) algorithm to predict the number of weeks an album spends on the Billboard chart `weeks_on_billboard` based on characteristics such as `release_year`, `peak_billboard_position`, and `spotify_popularity`. This model could help music industry analysts, artists, and record labels understand the factors that contribute to an album's longevity on the chart and make informed decisions about marketing and production strategies.

Our current model has a prediction error, as measured by the Root Mean Squared Error (RMSE), of approximately 74.71 weeks. This model has room for improvement, given that the `weeks_on_billboard` values has an Interquartile range of about 50 weeks.

Introduction

The data used to build this model comes from Rolling Stone’s rankings dataset, which includes information about album releases, chart performance, and artist details. The dataset was sourced from TidyTuesday, compiles information from Rolling Stone magazine’s renowned “The 500 Greatest Songs of All Time” lists. These lists were published in 2003, 2012, and 2020, each representing a snapshot of the most influential and celebrated songs according to music critics, artists, and industry experts at the time.

Variable	Class	Description
<code>sort_name</code>	character	Name used for sorting
<code>clean_name</code>	character	Clean name
<code>album</code>	character	Album name
<code>rank_2003</code>	double	Rank in 2003. NA if album not released yet or not in top 500
<code>rank_2012</code>	double	Rank in 2012. NA if album not released yet or not in top 500
<code>rank_2020</code>	double	Rank in 2020. NA if not in top 500
<code>differential</code>	double	2020-2003 Differential. Negative value if it went down in the chart. Positive value if it went up.
<code>release_year</code>	double	Release Year
<code>genre</code>	character	Album Genre
<code>type</code>	character	Album Type
<code>weeks_on_billboard</code>	double	Weeks on Billboard
<code>peak_billboard_position</code>	double	Peak Billboard Position
<code>spotify_popularity</code>	double	Spotify Popularity. NA if not on Spotify
<code>spotify_url</code>	character	Spotify URL. NA if not on Spotify
<code>artist_member_count</code>	double	Number of artists in the group

Variable	Class	Description
artist_gender	character	Gender of the artist(s). Male/Female if it's a mixed-gender group
artist_birth_year_sum	double	Sum of the artists' birth years
debut_album_release_year	double	Debut Album Release Year
ave_age_at_top_500	double	Average age at top 500 Album
years_between	double	Years Between Debut and Top 500 Album
album_id	character	Album ID. NOS at the beginning of the ID if not on Spotify

Part of the dataset shows the album rankings on the Billboard 200 Chart. The latter is a widely recognized measure of an album's commercial success. Understanding the factors that contribute to an album's longevity on the chart can provide valuable insights for artists, record labels, and music industry analysts. For example:

Artists: Can use this information to plan album releases and marketing strategies.

Record Labels: Can identify trends and allocate resources effectively.

Analysts: Can predict the potential success of new albums based on historical data.

In this project, we aim to answer the following question: Can we use characteristics of an album (e.g., release year, peak chart position, Spotify popularity) to predict how long it will stay on the Billboard chart?

To answer this question, we used a k-nearest neighbors (kNN) regression model and data from Rolling Stone's rankings. We found that our model has a prediction error of approximately 74.71 weeks, which is reasonable given the variability in the data. However, there is room for improvement, as the model explains 13.95% of the variability in weeks_on_billboard.

Methodology

Importing the data

We imported the data set from TidyTuesday's Github into the data directory. Using domain expertise and exploratory analysis, we chose to focus on the following predictor variables:

Table 1. Predictor variables used for analysis, chosen for their utility by domain expertise.

Predictor	Description
release_year	The year the album was released.
peak_billboard_position	The highest position the album reached on the Billboard chart.
spotify_popularity	A measure of the album's popularity on Spotify (0–100 scale).
artist_gender	The artist's gender.

Preliminary Analysis

Loading The Necessary Libraries

The necessary packages and libraries from the R programming language used in this analysis include `tidyr`, `dplyr`, `readr`, `ggplot2`, `ggpubr` and `caret`.

Wrangling

The data set used in this project was cleaned and prepared for analysis through several steps. First, numerical columns which had a large number of missing values (25% of total) and were poorly correlated as found in the EDA later were removed. Categorical variables like `sort_name`, `clean_name`, `album(name)`, `spotify_url`, etc. were removed, as they were not relevant to the analysis, and it would be difficult to use these variables as they are close to unique identifiers. Next, missing values were handled: the `genre` and `type` columns were imputed using the most frequent strategy, and the `weeks_on_billboard` column was imputed using the mean strategy, as removing these would have significantly skewed the analysis. Rows with missing values in other columns were dropped to ensure a complete dataset. Finally, all numeric predictors were scaled to ensure equal weighting in the kNN algorithm. These steps resulted in a clean, analysis-ready dataset.

Data Splitting

The data was partitioned into training (80%) and testing (20%) sets. The relationship between `weeks_on_billboard` and the predictor variables was visualized using scatter plots and boxplots to assess their utility for predictive modeling. Correlation was also used to assess the relationship between the response and the predictors.

Exploratory Analysis

In the exploratory data analysis (EDA), we first examined relationships between numeric variables and `weeks_on_billboard` using scatter plots with linear trendlines. This helped us visualize potential linear associations and identify predictors that might be strongly related to the target variable. Correlation values were also computed to quantify the strength of these relationships, highlighting which variables have positive or negative impacts on Billboard longevity.

Furthermore, we explored categorical variables — `genre`, `artist_gender`, and `type` — through boxplots to assess how `weeks_on_billboard` varies across different categories. These visualizations provided insights into how factors like album type and artist gender may influence chart success, and revealed potential group-level differences that could be important for modeling. Overall, the EDA gave a clearer picture of key patterns and relationships in the data.

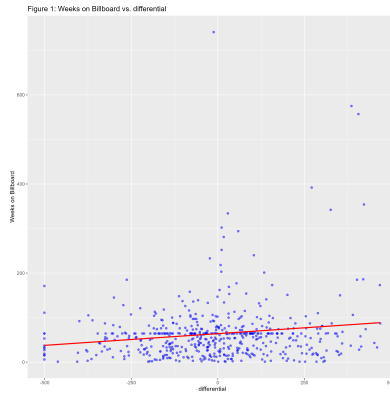


Figure 1: Weeks on Billboard Vs Change in Album's Ranking from 2003 to 2020

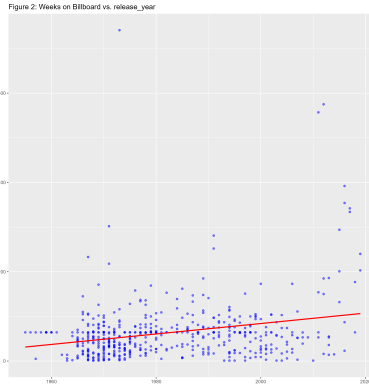


Figure 2: Weeks on Billboard Vs Release Year

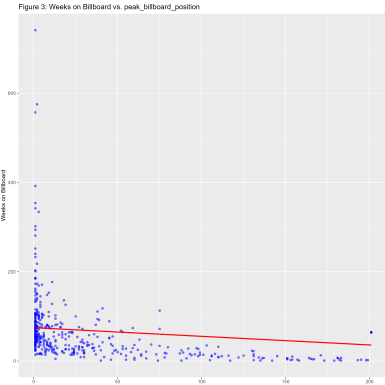


Figure 3: Weeks on Billboard Vs Peak Billboard Position

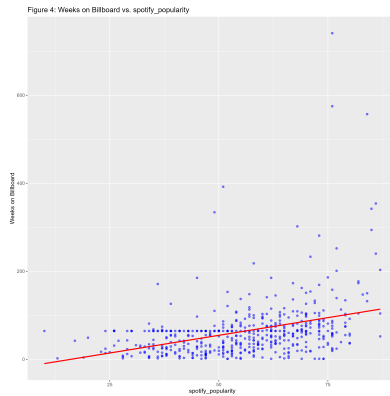


Figure 4: Weeks on Billboard Vs Spotify Popularity

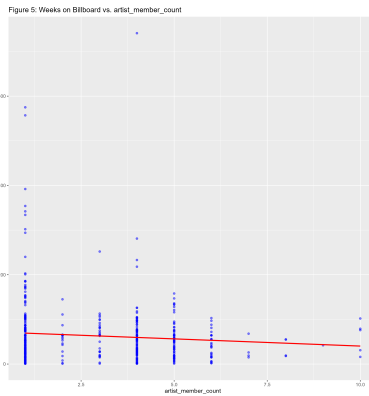


Figure 5: Weeks on Billboard Vs Artist Member Count

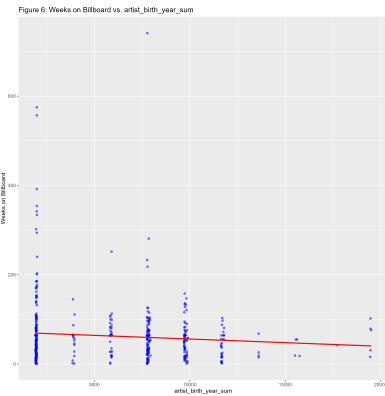


Figure 6: Weeks on Billboard Vs Artist Birth Year Total

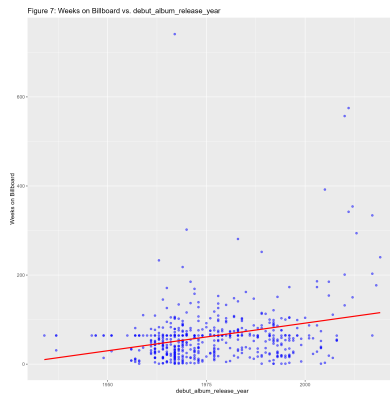


Figure 7: Weeks on Billboard Vs Debut Album Release Year

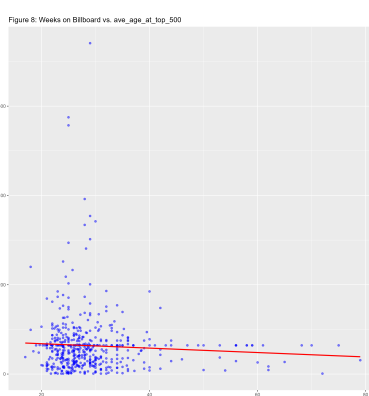


Figure 8: Weeks on Billboard Vs Average Age

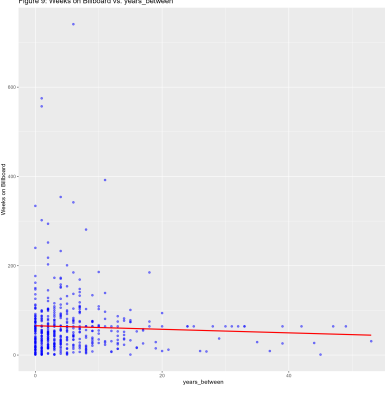


Figure 9: Weeks on Billboard Vs Age of the Album Relative to the Debut Album

Predictor Analysis

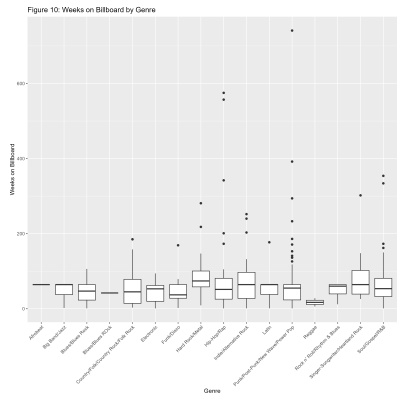


Figure 10: Weeks on Billboard By Music Genre of Album

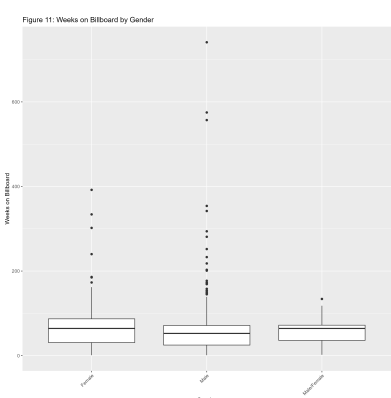


Figure 11: Weeks on Billboard By Gender

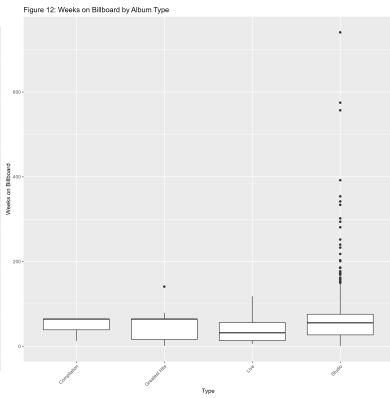


Figure 12: Weeks on Billboard By Music Presentation

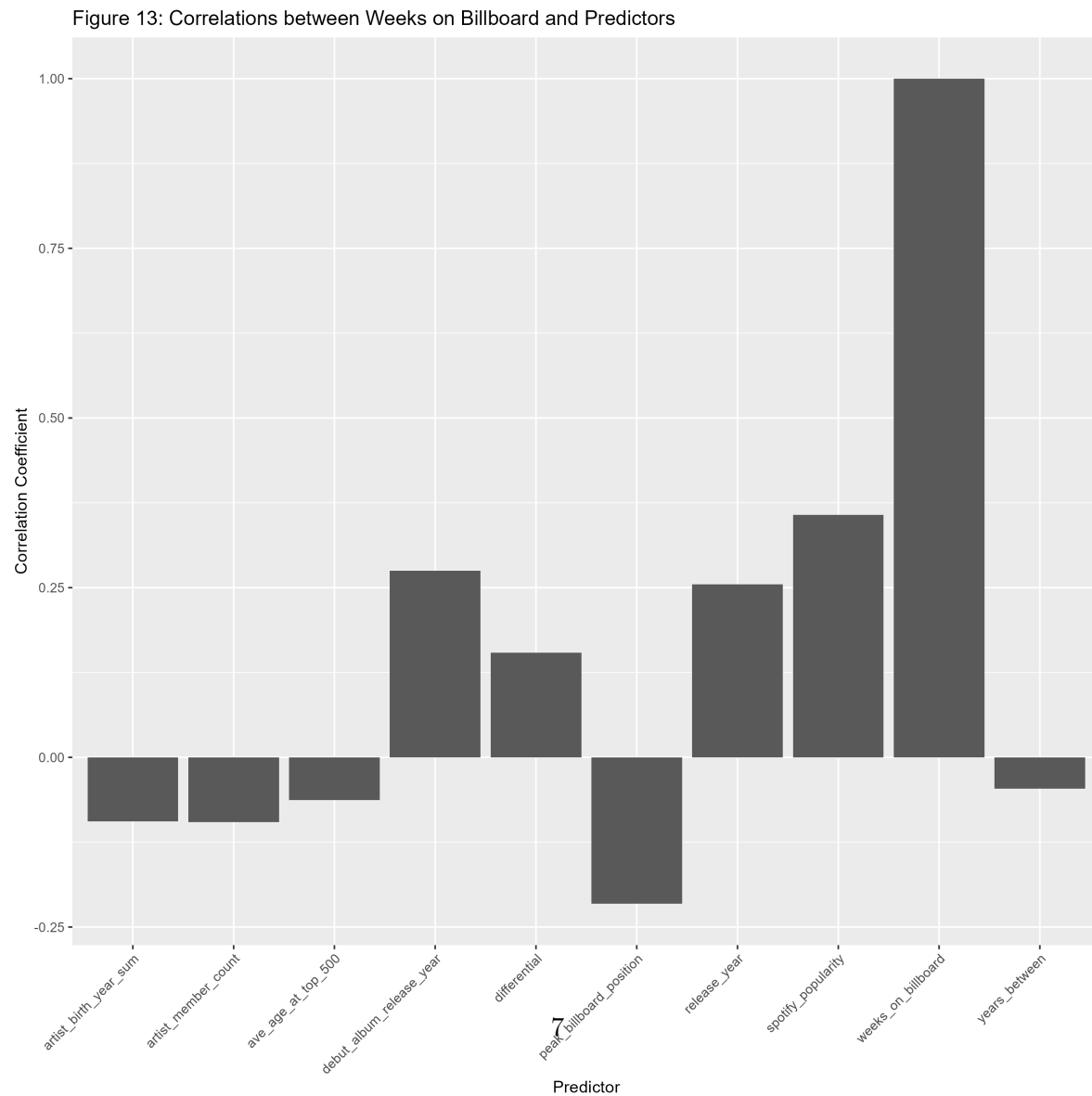


Figure 13: Predictor Barplot

Predictor Selection and Model Execution

For predictor selection, we focused on identifying variables that had meaningful relationships with `weeks_on_billboard` and were relevant to understanding album success. Variables with an absolute correlation greater than 0.2 were prioritized, as these are more likely to contribute valuable information to the model. Based on this criterion, we selected `release_year`, `peak_billboard_position`, `spotify_popularity`, and `debut_album_release_year` as predictors. Variables with weak correlations, such as `differential`, `artist_member_count`, and `artist_birth_year_sum`, were excluded to prevent unnecessary complexity and reduce the risk of overfitting.

After selecting predictors, we cleaned and scaled the datasets using min-max normalization to ensure all numeric variables were on a comparable scale. We also handled mismatches in categorical levels between training and testing datasets by filtering out unseen categories from the test set. A k-Nearest Neighbors (kNN) model was then trained using 5-fold cross-validation to tune the optimal value of k, evaluating performance with RMSE. The final model was assessed on the test set, where both RMSE and R-squared were calculated to measure prediction accuracy and explained variance. This process ensured a robust approach to selecting meaningful predictors and building a well-tuned predictive model.

kNN Model Tuning Results with Best k: 12
RMSE on Testing Set: 74.7097745978531 R-squared on Testing Set: 0.139515103258983

k	RMSE	Rsquared	MAE	RMSESD	RsquaredSD	MAESD
1 1	78.0092348473616	0.106972862075531	40.8267875073844	27.6401175509751	0.0798483388179309	8.66450100856507
2 2	65.6827668695226	0.146519849969749	36.6827651088228	22.3388064176595	0.0798231349642107	7.75511785459958
3 3	64.6864558143401	0.116901673586462	36.8650618612157	17.747797172946	0.0290221772987358	5.16717575633601
4 4	60.7047077151264	0.18272529598412	34.5509288628039	14.3685385131318	0.0976958794493373	3.94027358904303
5 5	60.8738758910583	0.168306869603794	34.7232716556947	15.2854014453066	0.0973950421460871	3.93990406413411
6 6	60.2823539182975	0.169415636511025	34.2793303781079	16.1996211428463	0.0937683443554025	4.08221390483195
7 7	60.2714630293188	0.165409172591062	34.6797970899293	16.5778771594567	0.0971562040745452	3.72618503460276
8 8	59.8780836451355	0.167857713587753	34.6423837754126	17.4899867303567	0.0731760949914562	3.62382118500529
9 9	59.8590333225524	0.162444561753753	34.5746056898942	17.9844111015912	0.0645594659701456	3.61555829923192
10 10	60.4199747974283	0.144461796798338	34.7839570391729	18.2968791048855	0.0594291992083143	3.34123836562299
11 11	60.0462206119883	0.150836509193393	34.5184996976605	18.5566033166414	0.0559606233034632	3.48708712094674
12 12	59.8182757718397	0.155700343886694	34.4467399431601	18.73321470725	0.0533869206505891	3.51571109757
13 13	59.8803921640638	0.152014192248085	34.4625840479259	18.8193523833303	0.05453619943672	3.38141841691406
14 14	60.2298216633704	0.140952902325044	34.837728279047	19.068221324572	0.0458872051464762	3.4361769315885
15 15	60.2310263245065	0.142359061325933	34.7845997342668	19.275942453316	0.0436015746309528	3.60014715381266
16 16	60.4965913542826	0.132390954732138	34.9536002099255	19.4268486634794	0.0392002293764242	3.83941992969662
17 17	60.5373377762632	0.133904980466283	34.9959044647534	19.6113425707627	0.0496380868929684	3.97340044708287
18 18	60.6113346870646	0.129774973554099	35.0850667972847	19.6783619045962	0.0471729592369782	4.01848432175445
19 19	60.785190481666	0.126895504219277	35.226744075341	19.9035354869533	0.0586405274862612	4.11311953541045
20 20	61.097305649152	0.117806623357344	35.5091042394724	19.8893433275952	0.0577901589554087	4.11150001396386

Figure 14: knn-tuning results

Discussion

Our interpretation of the data: The correlations between the predictors and the weeks on the billboard were generally weak. The variable that correlated with the weeks on the billboard most was the spotify popularity, which is expected since both are direct measurements of the popularity of the album. The correlation was 0.36. The variable that correlated with the weeks on the billboard the least was the years between the release date of the album and the first album the artist released. This was a bit surprising since people might expect more experienced artists to make more popular works. The correlaton was -0.05.

Implications of the data: The data could be used to predict the peak productivity of artists and determine which artists show the most promise in regard to their future career opportunities. Our data seems to imply that artists peak early in their careers. This may imply that the conditions required to produce popular music are mainly circumstantial and are difficult to recreate. This may raise future questions such as whether hiring based on experience is a good practice in some industries.

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